VIGIL: An Interpretable Machine Learning Pipeline for Deforestation Risk Prediction in Vietnam at 1 km Resolution

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***Abstract*— Deforestation monitoring requires both reliable historical mapping and actionable risk signals that support proactive intervention. VIGIL's objective is a nationwide deforestation early-warning capability for Vietnam: a complete pipeline that acquires data to train interpretable machine-learning models and deploys their outputs on an interactive public platform with early-warning alerts for forest rangers. A model general enough for nationwide use requires three data pillars - geographic, climatic, and human/production-context (local forest-management and logging records that distinguish licensed timber harvest from illegal clearing in the ground-truth label). The human/production pillar is not yet built nationwide: it requires forest-management data from local governments across every province, which Vietnam's Ministry of Agriculture and Environment (MAE) is compiling into a nationwide Forest Resource Monitoring System (FRMS); MAE has approved the team's use and publication of research based on this system, but the dataset remains under preparation and has not yet been delivered. The team therefore first builds and validates the pipeline on the two pillars already open and accessible - geographic and climatic - using public 1 km grid-cell predictors from Hansen Global Forest Change, SRTM terrain, CHIRPS rainfall, and Sentinel-2 optical bands and indices (2023). We report this pipeline's first end-to-end validation on Gia Lai Province, whose provincial government has already cross-checked the resulting Hansen-derived labels against its own local forest-management records - the only province where this validation currently exists. Developed on K'Bang District and evaluated for geographic transferability on Mang Yang District, logistic regression and random forest substantially outperform a baseline score, achieving AUC of 0.817 and 0.861 on the provided split, and 0.589 and 0.535 under cross-district transfer. VIGIL's province-agnostic public platform (https://www.vigil.green/) already deploys these risk maps for Gia Lai alongside a crowdsourced field-reporting channel, providing an operational foundation for nationwide extension as local validation data becomes available elsewhere.**

Keywords— deforestation; Vietnam; nationwide early-warning; Gia Lai pilot; VIGIL; risk mapping; interpretable machine learning; logistic regression; random forest; Sentinel-2; crowdsourced monitoring

I. INTRODUCTION

Vietnam's forest landscapes deliver critical ecosystem services, yet forest loss remains persistent where land-use pressure, accessibility, and governance constraints interact; most monitoring is reactive, recording loss only after it occurs. VIGIL's objective is a complete, nationwide deforestation early-warning pipeline: acquiring data to train interpretable machine-learning models, then deploying their outputs on an interactive public platform with early-warning alerts for forest rangers. A model general enough for nationwide use requires three data pillars - geographic, climatic, and human/production-context - the last capturing local forest-management and logging records so that licensed timber harvest is not mislabeled as illegal deforestation in the ground-truth signal. Building this human/production pillar at national scale requires forest-management data from local governments across every province; Vietnam's Ministry of Agriculture and Environment (MAE) is compiling this into a nationwide Forest Resource Monitoring System (FRMS) and has approved the team's use and publication of research based on it, but the dataset remains under preparation and has not yet been delivered.

The team therefore first builds and validates the pipeline using the geographic and climatic pillars alone, since these sources are open and immediately accessible - developing the full chain from data integration through public deployment, and cross-checking the resulting labels against one province's local forest-management records where that validation was already possible (Section V). Our contributions are: (1) a general grid-based pipeline integrating open geographic and climatic predictors with interpretable machine-learning models; (2) baseline-versus-machine-learning comparisons under a provided split and a cross-district transfer test; (3) capture-rate metrics translating model rankings into patrol priorities; (4) explainability outputs suitable for stakeholder communication; (5) VIGIL's public web platform and crowdsourced field-verification channel, both designed to be province-agnostic; and (6) a concrete path to nationwide

coverage once the human/production-context pillar is available everywhere (Section VI).

II. Problem Statement

Applied risk-mapping studies often lack end-to-end reproducibility, do not test geographic transferability, and produce outputs that are difficult for non-technical decision makers - such as park rangers - to act on. In many provinces, forest-protection resources are limited relative to the area under surveillance, and satellite-derived loss labels alone cannot distinguish licensed production logging from illegal clearing - precisely the gap the human/production-context pillar (Section I) is intended to close once local forest-management data is available nationwide. VIGIL uses a regular 1 km grid because it is interpretable for patrol planning and standardizes predictors and outcomes across data sources, regardless of region.

The core problem addressed in this paper is therefore whether a model trained with only open geographic and climatic data on one district produces risk rankings that remain useful - better than a naive baseline - on a different district it has never seen, using only the data pillars already available today. Section V evaluates this question on the one province where local government cross-checking of the resulting labels was already possible.

III. Proposed Solution

A. Data Sources

All inputs are public datasets processed in Google Earth Engine, with derived tables and figures produced in Python. Table I summarizes the four data sources. NDVI and NBR are computed from Sentinel-2 as (B8−B4)/(B8+B4) and (B8−B12)/(B8+B12); rainfall predictors combine a 12-month total and a 2000-2024 climatology from CHIRPS. Forest-loss labels come from Hansen lossyear (any loss 2001-2024), restricted to cells with baseline canopy cover ≥ 30%. These four sources jointly form the geographic and climatic pillars introduced in Section I; the human/production-context pillar - local forest-management and logging-permit records - requires data from every province's local government, which the Ministry of Agriculture and Environment is compiling nationally and has not yet delivered (Section VI).

TABLE I. SUMMARY OF DATASETS INTEGRATED IN THE PIPELINE

Dataset	Layers/Variables	Native res.	Time span	Role
Hansen GFC (v1.12)	treecover2000; lossyear-derived labels	30 m	2000 baseline; loss 2001–2024	Forest baseline + historical loss label
SRTM	mean_elevation_m; mean_slope_deg	30–90 m	Static	Terrain predictors
CHIRPS	rain_last12m_total_mm; rain_mean_annual_2000_2024_mm	~5 km	2000–2025	Rainfall predictors
Sentinel-2 SR (2023 composite)	B2, B3, B4, B8, B11, B12; NDVI; NBR	10–20 m	2023	Optical predictors (vegetation and disturbance signatures)

B. Grid-Based Modeling Architecture

Each district is tiled into a 1 km grid; outcome labels and predictors are aggregated per cell in GEE. The master table supports two validations: a provided train-test split, and a cross-district transfer test training on all cells of one district and testing on all cells of the other. Sampling is balanced by design, which stabilizes fitting but does not reflect natural prevalence, so probability calibration is recommended before full-grid deployment.

C. Predictive Algorithms

We compare a transparent baseline (a standardized-predictor z-score), logistic regression (interpretable, directionally-signed coefficients), and random forest (captures nonlinear interactions and yields feature importance). Performance is reported via AUC, average precision, thresholded precision/recall/F1, and capture-rate - the fraction of positive test cells contained in the top 5% and 10% highest-risk cells - translating model rankings into patrol priorities.

IV. Implementation

A. Modeling Pipeline

GEE scripts handle data access, masking, and aggregation; Python scripts train models, compute metrics, and export figures and maps, with intermediate tables saved for independent verification.

B. VIGIL Web Platform

VIGIL's public platform (https://www.vigil.green/) is a Next.js/TypeScript frontend rendering an interactive MapLibre GL JS map, backed by a FastAPI service (SQLAlchemy/GeoAlchemy2) over PostgreSQL + PostGIS, containerized with Docker and served via Nginx/PM2. Its architecture is province-agnostic: it renders grid-cell risk scores for whichever region has been modeled, with dark, light, and satellite basemaps and real-time filters - a 2001-2024 time slider and sliders for slope, elevation, tree cover, and rainfall. Section V shows the platform's first deployed instance and its operational warning zone, derived using a best-F1 threshold; displayed metrics for that instance are being reconciled with the verified values reported there.

C. Crowdsourced Field Verification

Visitors can submit a geolocated field report, optionally with a photograph, which is moderated and rendered as a map marker after approval. Reporter IP addresses are salted and SHA-256-hashed rather than stored in plaintext, and submissions are rate-limited (3 per 10 minutes per IP). The platform currently shows static risk scores rather than the explainability outputs in Section V.B, and does not yet feed validated field reports back into retraining - both are priorities

for future work (Section VI), alongside extending local forest-management cross-checking, of the kind described in Section V, to additional provinces as their data becomes available.

V. Results

We validate the pipeline described above on Gia Lai Province. Gia Lai's provincial government has already cross-checked the resulting Hansen-derived labels against its own local forest-management records, distinguishing licensed production logging from illegal clearing within the province - the only province where this local validation currently exists, and therefore the practical choice for this first end-to-end test, from data integration through public deployment. VIGIL is developed on K'Bang District and evaluated for geographic transferability on Mang Yang District; the two districts were chosen for contrasting forest and terrain conditions so that this pilot already probes the kind of variability a national rollout must handle. The master grid-cell table contains 6,000 balanced cells (K'Bang: 4,000; Mang Yang: 2,000), split into 4,788 training and 1,212 test cells.

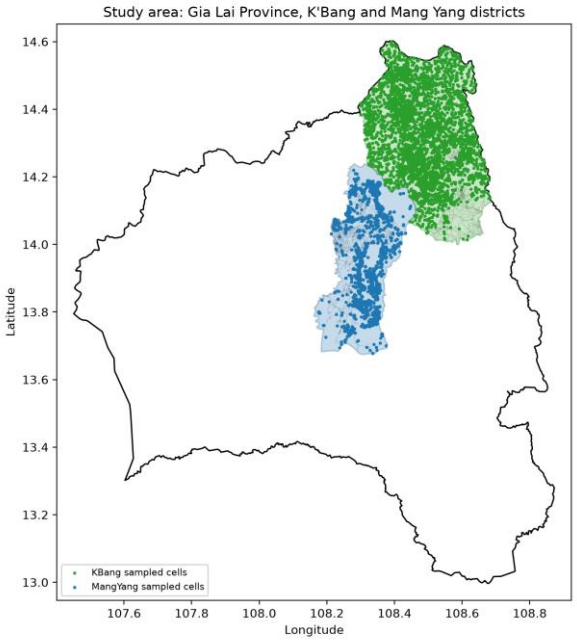


Fig. 1. Study area: Gia Lai Province, K'Bang and Mang Yang districts, and sampled grid-cell locations.

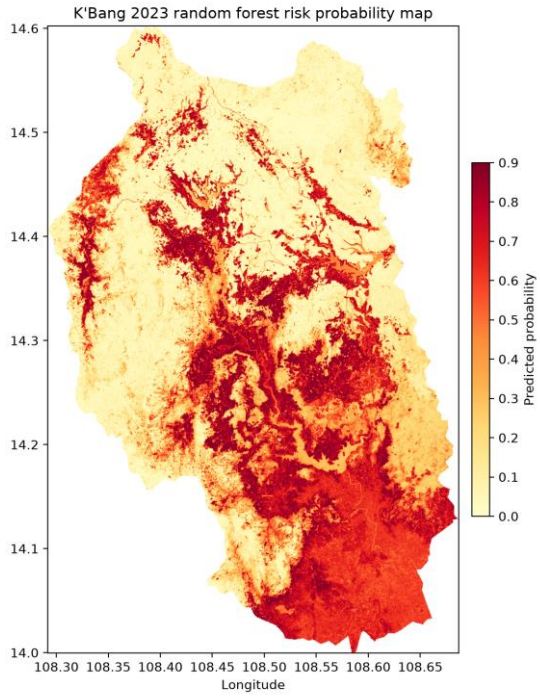


Fig. 2. K'Bang 2023 predicted deforestation-risk probability map, as served by the VIGIL platform's first deployed instance.

A. Baseline vs. Machine Learning Performance

Table II reports model performance for the provided split and the cross-district transfer test, transposed (metric × model) so each half fits within one column. On the provided split, both logistic regression and random forest substantially improve discrimination relative to the baseline. Transfer performance decreases, consistent with geographic domain shift, but still provides a prioritization signal. A sensitivity check using an alternative Hansen-loss label definition yielded consistent conclusions (full results omitted for space).

TABLE II(A). MODEL PERFORMANCE — PROVIDED SPLIT

Metric	Baseline	LogReg	RF
AUC	0.504	0.817	0.861
AP	0.559	0.815	0.841
Precision	0.493	0.715	0.782
Recall	0.489	0.835	0.794
F1	0.491	0.771	0.788
Cap@5%	0.077	0.089	0.087
Cap@10%	0.129	0.179	0.176
N_test	1212	1212	1212
N_pos	620	620	620

TABLE II(B). MODEL PERFORMANCE — AREA TRANSFER (K→M)

Metric	Baseline	LogReg	RF
AUC	0.376	0.589	0.535
AP	0.416	0.582	0.533
Precision	0.495	0.505	0.515
Recall	0.962	0.825	0.698
F1	0.654	0.627	0.593
Cap@5%	0.028	0.062	0.059

Cap@10%	0.059	0.135	0.111
N_test	2000	2000	2000
N_pos	1000	1000	1000

B. Curves and Explainability

Figure 3 shows ROC curves for the provided split; Figure 4 shows random forest feature importance. Logistic regression coefficients (not shown) indicate that higher rainfall and lower elevation/slope are associated with increased risk, consistent with the feature-importance ranking.

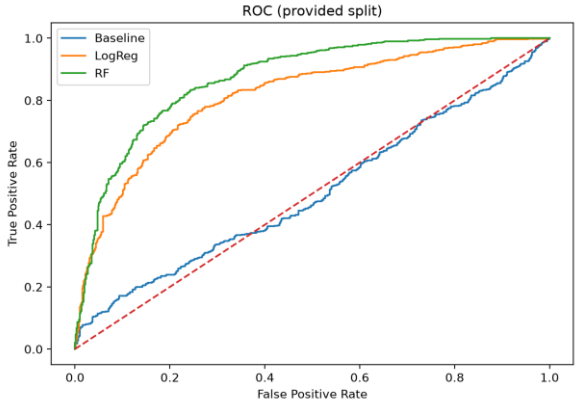


Fig. 3. ROC curves for the provided train-test split.

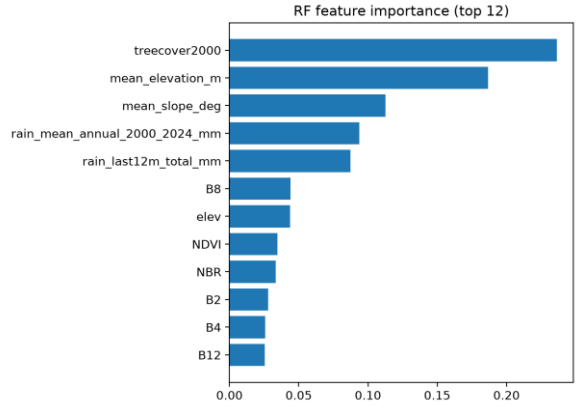


Fig. 4. Random forest feature importance.

VI. Discussion

VIGIL's logistic regression and random forest substantially outperform the baseline on the provided split, showing that terrain, rainfall, and optical predictors carry meaningful signal. Transfer performance decreases when trained on K'Bang and evaluated on Mang Yang, consistent with geographic domain shift from differences in land use, road access, and enforcement; capture-rate metrics remain useful here because they measure whether the model prioritizes a disproportionate share of positives even under shift. Independent, near-real-time Global Forest Watch alert layers complement VIGIL's predictive risk maps rather than duplicating them, since alerts flag loss that has already started.

Explainability pairs directional logistic-regression coefficients with random forest feature importance for stakeholder-facing communication, though neither implies causality. Because training samples are balanced by design, risk probabilities should be recalibrated before full-grid operational use. Future work should add time-based validation using loss_first_year, incorporate road-distance and land-tenure predictors, surface explainability directly on the VIGIL map, and connect crowdsourced field reports (Section IV.C) back into periodic retraining.

Path to Nationwide Scale

VIGIL's objective is nationwide coverage; Gia Lai served as the first pilot because it is the only province where local government cross-checking of the resulting labels against forest-management records was already possible (Section V). Extending beyond this pilot requires that same local validation, and the broader human/production-context data pillar it feeds (Section I), from every other province. Vietnam's Ministry of Agriculture and Environment (MAE) is compiling this into a nationwide Forest Resource Monitoring System (FRMS) and has approved the team's use and publication of research based on it, but the dataset remains under preparation and has not yet been delivered; this pilot was deliberately not delayed pending that delivery.

Independent of that delivery, the approach scales along four layers: Data - the GEE/Python pipeline needs only a new administrative boundary and grid per province, since every geographic and climatic input is public and nationwide; Modeling - rather than one national model, the baseline/LogReg/RF comparison should be repeated per region (extended to boosting methods such as XGBoost and LightGBM), selecting whichever generalizes best locally; Validation - each additional province needs its own local government cross-check of Hansen-derived labels against forest-management records, as already completed for Gia Lai, before FRMS can provide an independent national ground-truth layer; Deployment - VIGIL's architecture is already province-agnostic, so scaling is a data-loading exercise rather than a system redesign.

VII. Ethical Considerations

VIGIL is intended to help rangers and provincial agencies prioritize limited patrol resources toward the highest-risk cells (Section V); it must not be used to justify punitive land seizures, displacement of communities, or automated enforcement without human verification - particularly outside Gia Lai, where local forest-management cross-checking has not yet confirmed that flagged loss excludes licensed production logging (Section VI). Because training samples are balanced by design and transfer performance degrades outside the training district (Section V.A), outputs should be treated as

a prioritization aid - recalibrated and locally re-validated before guiding binding decisions - not as proof of wrongdoing at any location.

The crowdsourced field-reporting channel protects reporter identity via salted, SHA-256-hashed IP addresses, rate-limits submissions to deter abuse, and moderates uploaded photographs and comments before public display.

VIII. Conclusion

We present VIGIL, a reproducible deforestation risk-prediction pipeline and public web platform designed for nationwide deployment across Vietnam, validated end-to-end on Gia Lai Province - the one province where local government cross-checking of the resulting labels was already possible - at 1 km resolution across two districts. Built on Hansen forest-loss history, SRTM terrain, CHIRPS rainfall, and Sentinel-2 optical features, VIGIL produces interpretable models and operational ranking metrics, and its live deployment shows these outputs can already be operationalized while highlighting remaining gaps - map-level explainability, a field-report feedback loop, and the local forest-management validation still needed from every other province. The Ministry of Agriculture and Environment has approved the team's use and publication of research based on its nationwide Forest Resource Monitoring System; the next step is the Ministry's delivery of that dataset, after which the four-layer scaling path in Section VI can be pursued directly.

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